

# tobler: Spatial interpolation and areal analysis for Python

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## Software

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## Summary

tobler is an open-source Python library for spatial interpolation and areal data transformation, designed to support a wide range of workflows in geographic information science (GIScience), spatial data science, and applied spatial analysis. The package provides a unified, extensible framework for transferring data between incompatible zonal systems—an essential task when working with spatially aggregated data such as census geographies, administrative boundaries, or service areas.

Areal interpolation is a foundational problem in spatial analysis, arising whenever data collected over one set of polygons (source zones) must be reallocated to another (target zones) with different spatial configurations. tobler implements a suite of methods for this task, ranging from simple area-weighted interpolation to more advanced dasymetric and model-based approaches. The package is designed to operate natively on GeoPandas GeoDataFrames, enabling seamless integration into modern Python-based geospatial workflows.

As part of the Python Spatial Analysis Library (PySAL) ecosystem (S. J. Rey et al., 2022; S. J. Rey & Anselin, 2007), tobler emphasizes composability, transparency, and reproducibility while providing robust implementations of areal interpolation methods for both applied and methodological research.

## Statement of need

Spatial data are commonly aggregated to administrative units such as census tracts, ZIP codes, or political boundaries, which often differ across datasets or change over time. As a result, transferring variables between incompatible zoning systems is a common challenge for longitudinal analysis, data integration, and cross-scale comparison, particularly in the social and behavioral sciences. While Python packages such as PyKriging (Murphy et al., 2025) and pyinterpolate (Moliński, 2022) support *continuous* spatial interpolation, they are not designed for interpolation between discrete zoning systems.

Traditional GIS software (e.g., ArcGIS, QGIS) provides tools for areal interpolation, but these implementations are often embedded in graphical interfaces, limiting reproducibility and automation. Moreover, they may not expose the full range of methodological options or allow for easy integration into data science pipelines.

Prior to tobler, the Python ecosystem support for areal interpolation was fragmented or limited. While foundational libraries such as GeoPandas provide data structures and geometric operations, they do not natively implement interpolation methods. As a result, users often

40 relied on *ad hoc* scripts or external tools, leading to inconsistent workflows and potential  
41 methodological errors.

42 `tobler` addresses these challenges by providing:

- 43     ▪ A coherent and well-documented API for areal interpolation
- 44     ▪ Native integration with GeoPandas and the PySAL ecosystem
- 45     ▪ Area-weighted, dasymetric, and model-based interpolation methods
- 46     ▪ Reproducible workflows

47 These capabilities make `tobler` particularly valuable for researchers and practitioners in fields  
48 such as geography, urban planning, public health, environmental science, and regional economics,  
49 where spatial data integration is a routine requirement.

## 50 State of the field

51 `tobler` is a component of the PySAL ecosystem (S. Rey, Stephens, et al., 2026), which  
52 provides a comprehensive suite of tools for spatial analysis in Python. This ecosystem consists  
53 of four layers of related packages:

- 54     ▪ `libpysal` for foundational spatial data structures and utilities (S. Rey, Wolf, et al., 2026)
- 55     ▪ `explore` for exploratory spatial analysis
- 56     ▪ `model` for spatial modeling
- 57     ▪ `viz` provides classification schemes for choropleth mapping

58 Within this ecosystem, `tobler` occupies a critical niche by providing the data integration and  
59 interpolation methods necessary to harmonize datasets before analysis in the `explore` or `model`  
60 phases.

61 `tobler` complements these packages by addressing the specific problem of spatial data  
62 transformation between incompatible zonal systems. Compared to desktop GIS platforms,  
63 `tobler` offers advantages like **Reproducibility**, **Transparency**, **Extensibility**, and **Integration**.

64 While similar functionality exists in other ecosystems (e.g., R packages such as `areal` (Prener  
65 et al., 2019) or `sf`-based workflows (Pebesma, 2018)), `tobler` provides a native solution for  
66 Python users.

## 67 Software design

68 `tobler` is designed with attention to both computational efficiency and usability. Spatial  
69 overlay operations, which are central to areal interpolation, can be computationally intensive  
70 for large datasets. The package leverages vectorized operations and efficient geometric libraries  
71 (via GeoPandas (Bossche et al., 2026) and Shapely (Gillies et al., 2026)) to handle these tasks.

72 The API design emphasizes clarity and consistency, with function signatures that explicitly  
73 distinguish between extensive and intensive variables. Intensive variables are things like rates or  
74 percentages that must be averaged when aggregating, whereas extensive variables are things  
75 like counts that must be summed during aggregation.

76 Additionally, `tobler` is developed with testing and documentation standards consistent with  
77 the Scientific Python ecosystem, ensuring reliability and maintainability.

## 78 Core Functionality

79 `tobler` organizes its functionality around several key interpolation paradigms, each  
80 corresponding to different assumptions about how variables are distributed within source zones.

## 81 Area-weighted interpolation

82 Area-weighted interpolation assumes that variables are uniformly distributed within each source  
83 zone and allocates values to target zones in proportion to the area of overlap.

84 `tobler` provides efficient implementations for both **extensive variables** (e.g., population counts)  
85 and **intensive variables** (e.g., rates or densities), ensuring appropriate handling of each type  
86 ([Goodchild & Lam, 1980](#)). The library also supports pycnophylactic adjustments to preserve  
87 totals where required ([Tobler, 1979](#)).

## 88 Dasymetric interpolation

89 Dasymetric interpolation uses ancillary data (e.g., land use or land cover) to model within-zone  
90 heterogeneity ([Eicher & Brewer, 2001](#); [Mennis & Hultgren, 2006](#); [Reibel & Agrawal, 2007](#)).  
91 `tobler` supports both vector- and raster-based dasymetric workflows, allowing users to integrate  
92 a wide range of auxiliary datasets. This is particularly useful in urban and environmental  
93 applications where fine-scale heterogeneity is important.

## 94 Model-based interpolation

95 Beyond deterministic approaches, `tobler` includes model-based methods that use statistical or  
96 machine learning techniques to estimate spatial distributions. These approaches incorporate  
97 covariates and capture more complex spatial patterns, providing improved accuracy in many  
98 contexts ([Flowerdew & Aitkin, 1982](#); [Flowerdew & Green, 1992](#)).

## 99 Integration with GeoPandas

100 All interpolation methods in `tobler` operate directly on GeoPandas GeoDataFrames and return  
101 GeoDataFrames, enabling seamless integration with the broader Python geospatial ecosystem.

## 102 Example workflow

103 A typical area-weighted interpolation in `tobler` is implemented as follows:

```
from tobler.area_weighted import area_interpolate

result = area_interpolate(
    source_df,
    target_df,
    extensive_variables=["population"],
    intensive_variables=["income"]
)
```

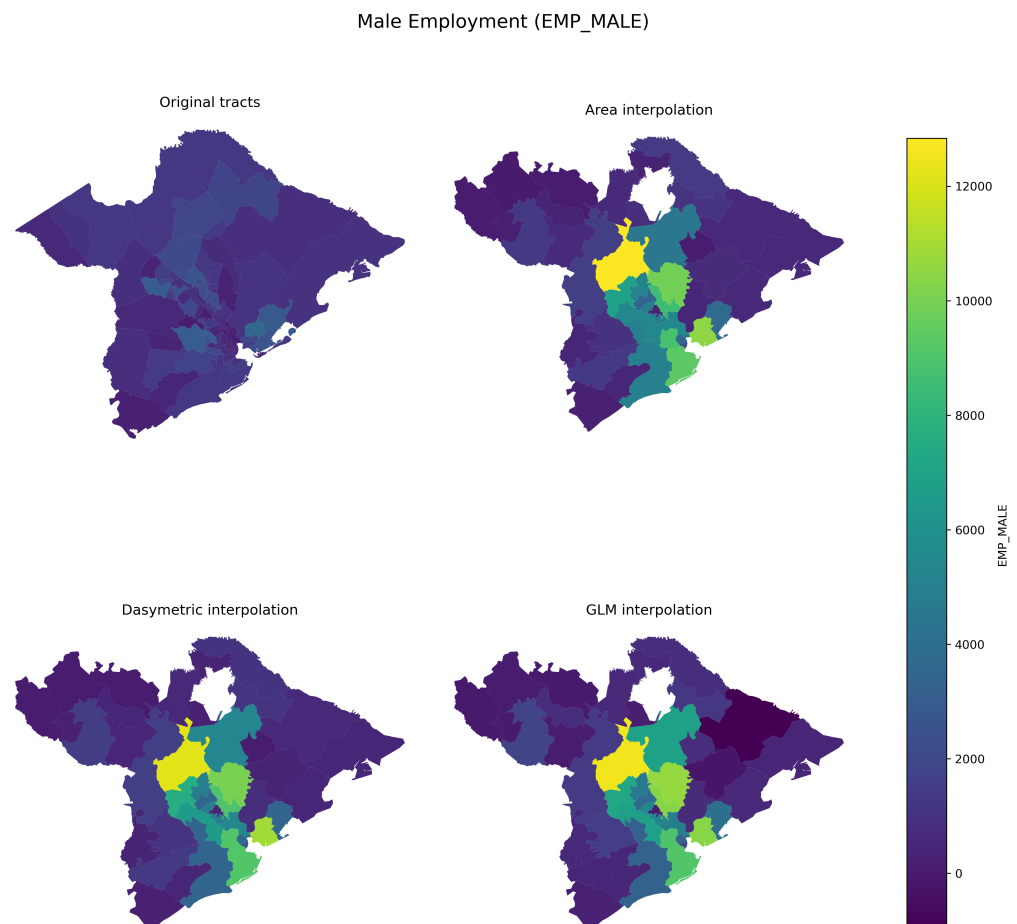
104 This operation transfers population counts and income measures from the source geometries  
105 to the target geometries, handling each variable type (extensive/intensive) appropriately.

106 When additional information about within-zone heterogeneity is available, dasymetric  
107 interpolation can be used to refine estimates. For example, population counts may be  
108 redistributed using a land cover raster to exclude uninhabited areas:

```
from tobler.dasymetric import masked_area_interpolate

result = masked_area_interpolate(
    raster="raster_file_name.tif",
    source_df,
    target_df,
    pixel_values = [21,22,23,24],
    extensive_variables=["population"]
)
```

109 Similarly, the user can execute a model-based approach using the `tobler.model.glm` function.  
110 [Figure 1](#) illustrates an example comparing interpolated values derived from different spatial  
111 configurations, highlighting how results may vary depending on the underlying geometry and  
112 interpolation approach.



**Figure 1:** Example of `tobler` usage for a single extensive variable – employed male population – (no intensive variable) in Charleston, SC, comparing census tracts and ZCTAs.

## Research impact statement

113 The package is actively used by the research community to transfer the data between various  
114 types of geographic boundaries. This is not limited to specific applications but covers use  
115 cases from continental analysis of emissions and health ([La Porta & Zapperi, 2024](#)), analysis  
116 of urban form and function ([Fleischmann & Arribas-Bel, 2022](#)), redistribution of census data  
117 to school districts for assessment of the Clean School Bus Rebate Program ([Osia et al., 2025](#)),  
118 quantification of radon exposure ([Lee et al., 2026](#)), and harmonization of vector and raster  
119 data for computer vision tasks ([Fleischmann & Arribas-Bel, 2024](#)).  
120

121 Moreover, the package is relied on in downstream software such as `atlasbr` for harmonization  
122 of Brazilian urban data ([Oliveira & Paiva Neto, 2025](#)), and is referred to in the `pygridmap`  
123 package by Eurostat ([Grazzini & Gaffuri, 2020](#)) as a reference implementation. The `tobler`  
124 package has made tangible contributions to spatial science, pedagogy, and applications in  
125 government and industry. In academia, the package is used as part of a data-processing pipeline

for research that examines the spatial-contextual influence on a variety of outcomes, including segregation (Wei et al., 2022), housing policy (S. J. Rey & Knaap, 2022), education policy (Osia et al., 2025; S. J. Rey et al., 2024), and pollution exposure (La Porta & Zapperi, 2024; Lee et al., 2026). It is also used in environmental science (Hu et al., 2023) and regionalization research (Feng et al., 2022).

In spatial data science education, *tobler* has become an integral part of many curricula. It is included in popular pedagogical resources, including two textbooks (Knaap, 2026; S. J. Rey et al., 2023), and is taught in graduate and undergraduate courses in universities across the globe, including the University of California (Berkeley, Irvine, Riverside, and Santa Barbara campuses), San Diego State University, Charles University, University of Liverpool, Northern Arizona University, and Temple University, among others.

In the public sector, the *tobler* package is used as part of a processing pipeline that powers urban planning and policymaking, including the *DemoLand* project from the Alan Turing Institute, developed as a part of the British National Land Data Programme (Geospatial Commission, 2023). The project piloted predictive modelling based on data interpolated to a unified hexagonal grid and is scaling the approach nation-wide.

## AI usage disclosure

No generative AI or LLMs were used for code production for *tobler* or the writing of this paper.

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